

PROJECT BASED LEARNING REPORT

ON

VerifyMate

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE FOR THE
PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING IN
INFORMATION TECHNOLOGY

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Sr. No. 27, Trimurti Chowk, Dhankawadi, Pune-43

AY 2024-25

Society for Computer Technology & Research's

PUNE INSTITUTE OF COMPUTER TECHNOLOGY

Sr. No. 27, Trimurti Chowk, Dhanakawadi, Pune-43

DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that the PROJECT BASED LEARNING report entitled

VerifyMate

submitted by

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is a bonafide work carried out by them in the second year of Engineering (AY 2024-25) under the supervision of **Mrs. Rachna Karnavat** and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University for the award of the Degree of Bachelor of Engineering (Information Technology).

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ABSTRACT

VerifyMate is a full-stack web application developed using HTML , CSS , JS , NODE , EXPRESS , MySQL, designed to streamline and automate the verification process of 7/12 land ownership documents, which are critical legal records in Indian land and property registration.

The system allows users to upload scanned copies or PDFs of 7/12 documents through a user-friendly web interface. On the backend, Tesseract OCR is integrated to perform high-accuracy text extraction from images or PDFs. To enhance the document processing pipeline, tools such as Sharp, PDF-lib, and Puppeteer are utilized for image preprocessing, PDF parsing, and format consistency. These technologies ensure that even low-quality or non-standard documents can be interpreted effectively.

Once the text is extracted, the system compares the retrieved information—such as survey number, landowner details, village name, and mutation entries—against verified datasets stored in MySQL. The verification logic is implemented using Node.js and Express, enabling secure API handling, asynchronous data processing, and real-time feedback. If the extracted values match the records in the database, the document is marked as "Verified." Otherwise, the system flags inconsistencies for manual review or rejection.

The entire verification workflow is built with scalability, efficiency, and security in mind. It removes the need for manual data entry and reduces the risk of document forgery or misrepresentation in property transactions.

By combining OCR-driven data extraction with intelligent rule-based comparison, VerifyMate serves as a reliable solution for property registration authorities, legal consultants, and individuals seeking to authenticate land records swiftly. The project showcases strong skills in full-stack development, file handling, backend logic, and real-world application of document processing technologies.

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LISTS OF ABBREVIATIONS

Sr. No.	Abbreviations	Full Form
1	JS	JavaScript
2	API	Application Programming Interface
3	IDE	Integrated Development Environment
4	OCR	Optical Character Recognition

INTRODUCTION

VerifyMate is a full-stack web application developed using **HTML, CSS, JavaScript, Node.js, Express, and MySQL**, focused on automating the verification process of **7/12 land ownership documents**—key legal records in Indian property registration. The platform enables users to upload scanned copies or PDFs of documents and uses **Tesseract OCR** for accurate text extraction. Advanced tools like **Sharp, PDF-lib, and Puppeteer** are used for preprocessing and format consistency. By comparing extracted data with verified entries in the database, VerifyMate ensures fast and reliable document authentication.

MOTIVATION

1. Manual verification of 7/12 land records is slow, error-prone, and vulnerable to forgery.
2. There's an increasing demand for secure and automated document verification systems in property-related processes.
3. Current systems lack integration of modern technologies like OCR and backend data matching.
4. Our goal is to streamline property verification, reduce fraud, and provide real-time trust to users, legal professionals, and authorities.
5. Solving this problem will help build transparency, accuracy, and efficiency in land transaction.

SCOPE

- Document Upload: Users upload scanned 7/12 property records (PDFs/images).
- OCR Extraction: Tesseract OCR extracts key details like survey number, owner name, and village name.
- Document Processing: Tools like Sharp, PDF-lib, and Puppeteer enhance readability and formatting.
- Database Matching: Extracted information is verified against authentic MySQL records.
- Real-Time Results: Documents are instantly marked as "Verified" or "Rejected".
- Manual Review: Mismatches or unrecognized data can be flagged for manual verification.
- User Interface: Clean, responsive, and intuitive web interface.
- Secure Backend: Built using Node.js and Express to ensure secure file handling and API operations.

OBJECTIVES

- **Automate the Verification Process:**

Develop a system that extracts data from scanned 7/12 property documents using OCR and verifies it against pre-stored authentic records, eliminating manual verification steps.

- **Ensure Accuracy and Efficiency:**

Use advanced document processing tools and intelligent matching algorithms to handle low-quality or inconsistent formats, ensuring accurate and fast results.

- **Build a Secure and Scalable Platform:**

Create a secure, user-friendly web application that can efficiently handle large volumes of document verifications and adapt to future enhancements or new document formats related to land records.

OUTCOMES

- **Reduced Manual Effort and Human Error:**

The system automates the verification process, minimizing the need for manual data entry and reducing the chances of mistakes.

- **Faster and More Reliable Document Verification:**

Real-time processing and accurate comparison with trusted records ensure quicker and more dependable verification of 7/12 property documents.

- **Improved Transparency and Trust in Land Transactions:**

By detecting inconsistencies and flagging potential fraud, the platform builds greater trust among users, legal authorities, and property stakeholders.

LITERATURE SURVEY

Sr. No	Title	Author	Year	Aim/Objective	Methodology	Gaps/Findings
1 (W)	Landeed: India's Property Title Search Engine	Landeed Team	2023	To provide users with instant access to land records such as 7/12, EC, ROR, etc.	Web-based platform using APIs to fetch data from state gov. portals	No support for direct OCR-based verification of uploaded documents
2 (A)	Surepass – Land Record Check API	Surepass.io	2022	To enable third-party services to verify land ownership using APIs	REST APIs integrated with government datasets and land record databases	Lacks document upload or scan verification, only verifies existing land record numbers
3 (W)	Property Title & Deed OCR Extractor	Affinda	2021	To extract data fields from scanned property documents using OCR	Uses Tesseract-based OCR with AI/ML models for field detection and parsing	Focuses on global documents, lacks specific support for Indian 7/12 extracts
4 (P)	An OCR-Based Document Verification System for Academic Records	Dr. B.K. Tripathi, Dr. Rakesh Kumar [IEEE Conference Publication]	2019	To verify academic certificates using OCR and database matching	Tesseract OCR, Python, MySQL database for extracted text comparison	Applies to academic records only, not land documents would be able

SYSTEM OVERVIEW

Front-end:

The front-end is the user-facing interface where users can upload scanned copies or images of their land records (7/12 documents). Built using modern web technologies like React and Tailwind CSS, it ensures a clean, responsive, and user-friendly experience across all devices. It displays document upload options, verification results (Verified/Rejected), and any manual review alerts.

Back-end:

The back-end is developed using Node.js and Express. It handles secure file uploads, manages API communication with OCR tools, processes documents using image/PDF libraries, and interfaces with the MySQL database. It also ensures the logic for real-time verification and manual review alerts is efficiently executed.

OCR Engine:

Tesseract OCR is used to extract text from the uploaded 7/12 documents. It identifies critical fields such as the survey number, owner's name, and village details which are necessary for validation.

Document Processing Tools:

Libraries such as **Sharp**, **PDF-lib**, and **Puppeteer** enhance document readability, standardize formats, and ensure clean extraction from various file types (PDFs, JPEGs, PNGs).

Database (MySQL):

The MySQL database stores the trusted and verified records (e.g., survey numbers, owner names, etc.). Extracted data is matched with these records to determine document authenticity.

Search Engine (Matching Logic):

A custom-built matching engine compares OCR-extracted data with database entries and classifies the result as "Verified" or "Rejected". Mismatches are flagged for manual review if needed.

IMPLEMENTATION DETAILS

The implementation of the document verification website involved several key components and considerations:

Frontend (HTML, CSS, JavaScript):

- The frontend was developed using HTML for structure, CSS for design, and JavaScript for interactivity.
- The UI is structured using modular sections like the home page, document upload section, and result display, promoting clarity and simplicity.
- Fetch API is used to communicate with backend endpoints, enabling smooth file submission and result retrieval.
- Responsive design principles were applied to ensure a consistent user experience across different devices. Responsive design principles were applied using CSS Flexbox/Grid and media queries to ensure the website works well across devices like phones, tablets, and desktops.
- Since no authentication is required, the website is kept simple and direct for public use without login.
- Basic client-side validation is implemented to ensure the uploaded file meets required formats (e.g., image or PDF).

Backend (Node.js):

- The backend was set up using Express.js, providing a foundation for routing and API endpoint management.
- Accept uploaded land record documents (7/12). Extract text using OCR. Compare extracted data with database records.
- **Tesseract.js**, an OCR (Optical Character Recognition) library, was integrated to extract meaningful text (e.g., owner name, village name) from uploaded images.
- Business logic was implemented to handle user requests, validate inputs interact with the database.
- Basic error handling and input validation were implemented to prevent crashes and guide the user in case of issues (e.g., unreadable image, unsupported file format).

Deployment and Infrastructure:

- The frontend and backend were deployed on separate servers or platforms, ensuring scalability and efficient resource management.
- Continuous Integration/Continuous Deployment (CI/CD) pipelines were set up to automate build, testing, and deployment processes.

PROJECT SCHEDULE

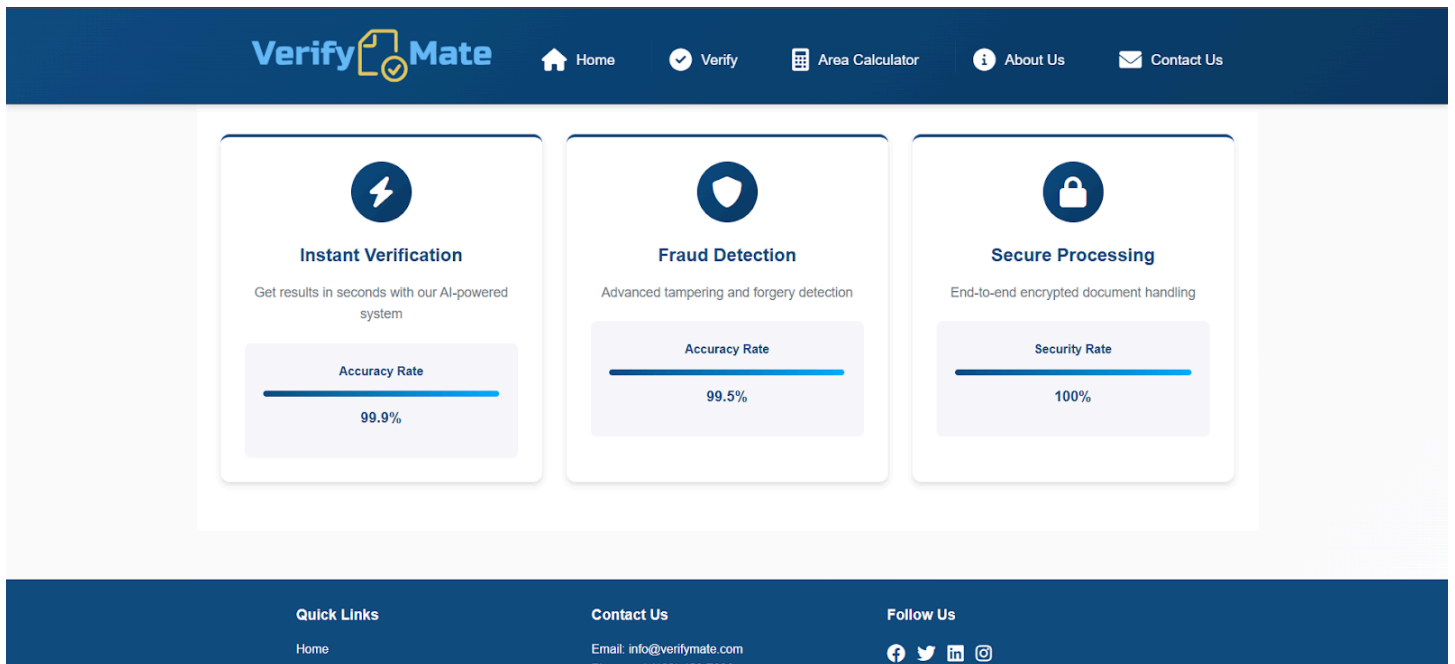
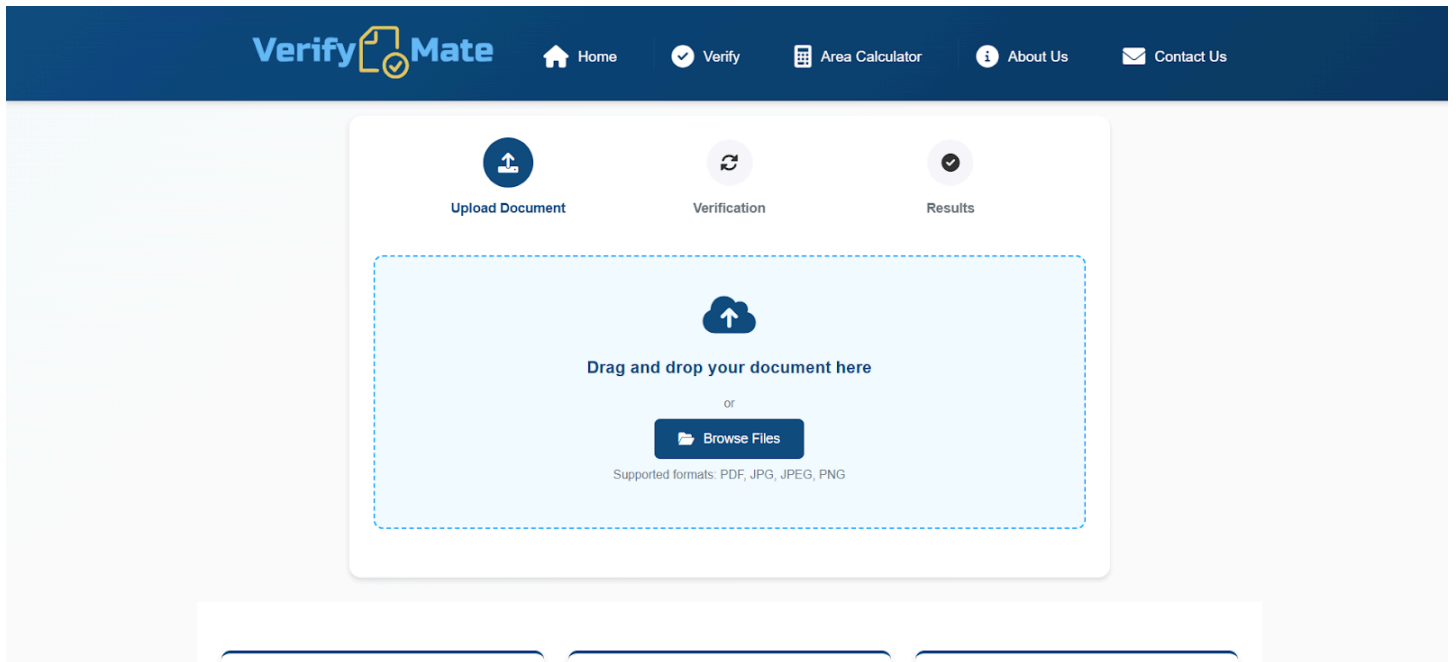
An elementary Gantt chart or Timeline chart for the development plan is given below. The plan explains the tasks versus the time (in weeks) they will take to complete.

	JAN		FEB				MAR				APR			
Requirement Gathering														
Analysis														
Design														
Coding														
Testing														
Implementation														
	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14

Table No.2 Project Scheduling

SCREENSHOTS

1. Verify Page



2.

Area Calculation

VerifyMate

[Home](#)

[Verify](#)

Area Calculator

[About Us](#)

[Contact Us](#)

Area Calculator

Calculate the area of your property quickly and accurately. Enter the dimensions below to get started.

Input Value

Enter value

From Unit

Square Meter (m²)

To Unit

Square Foot (ft²)

Result: -

Remove

+ Add Another Conversion

Total Sum: 0.000000

Quick Links

[Home](#)

[Verify](#)

[Area Calculator](#)

[About Us](#)

[Contact Us](#)

Contact Us

Email: info@verifymate.com

Phone: +1 (123) 456-7890

Address: 123 Main St, City, Country

Follow Us

[Facebook](#)

[Twitter](#)

[LinkedIn](#)

[Instagram](#)

3.

Contact Us Page



[Home](#)

[Verify](#)

[Area Calculator](#)

[About Us](#)

[Contact Us](#)



Our Location

123 Main Street, City,
Country



Phone Number

+91 1234567890



Email Address

info@verifymate.com

Send Us a Message

Fill out the form below and we'll get back to you as soon as possible

Your Name

Email Address

Subject

Your Message

Send Message

CONCLUSION

The Document Verification Website project has successfully met its objectives of providing a streamlined platform for users to upload, verify, and manage critical documents like property ownership records. The project leveraged HTML, CSS, JavaScript, and Node.js to build a robust, scalable, and efficient system.

The website features an intuitive interface, where users can upload scanned documents, which are processed through Optical Character Recognition (OCR) to extract key details. These details are then verified against the database, with real-time results provided to the user. The secure backend infrastructure ensures that user data is safely processed and stored. The responsive design ensures a smooth experience across various devices, making the platform easily accessible.

The implementation of Tesseract OCR, Sharp, PDF-lib, and Puppeteer helped in ensuring document readability and accuracy. The backend, built using Node.js with Express, handles the verification and business logic, while a secure system architecture protects sensitive data.

Looking ahead, the project has the potential for further improvements. By incorporating feedback from users, expanding its features, and optimizing for better scalability, the system can handle additional document types and more traffic in the future.

Overall, the Document Verification Website serves as an effective tool for document verification and provides a solid foundation for future enhancements. The combination of Node.js for the backend and modern frontend technologies has proven to be an effective solution in delivering a secure, user-centric application.

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